Accepted Manuscript



An update on metabolic syndrome: Metabolic risk markers and adipokines in the development of metabolic syndrome

Reena Kumari, Sandeep Kumar, Ravi Kant

PII: S1871-4021(19)30314-5

DOI: https://doi.org/10.1016/j.dsx.2019.06.005

Reference: DSX 1437

To appear in: *Diabetes & Metabolic Syndrome: Clinical Research & Reviews*

Received Date: 8 May 2019

Accepted Date: 7 June 2019

Please cite this article as: Kumari R, Kumar S, Kant R, An update on metabolic syndrome: Metabolic risk markers and adipokines in the development of metabolic syndrome, *Diabetes & Metabolic Syndrome: Clinical Research & Reviews* (2019), doi: https://doi.org/10.1016/j.dsx.2019.06.005.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Review Article

An update on metabolic syndrome: Metabolic Risk markers and adipokines in the development of metabolic syndrome.

Reena Kumari[1] Sandeep Kumar[2] and Ravi Kant[2]

1. Department of Biochemistry, King George's Medical University, Lucknow

2. Department of Molecular Biology AIIMS, Rishikesh

Address to Correspondence
Dr. Sandeep Kumar
Demonstrator
Department of Molecular Biology
AIIMS, Rishikesh
Sschaudhary55@gmail.com
9453440278

Conflict of Interest:- None

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

An update on metabolic syndrome: Metabolic Risk markers and adipokines in the development of metabolic syndrome

Abstracts-

Background:- Metabolic syndrome is a collection of physiological and biochemical abnormalities about 20-25 % of adult population in developing countries is suffering from metabolic syndrome. Previous research demonstrated that adipose tissue plays an important role in energy regulation via endocrine, paracrine and autocrine signals as results of obesity due to accumulation of adipose tissue to excess that by time affects negatively both physical and psychological health and well being, it has been found that adipose tissues produces a variety of factors known as ***"adipokines"*** which play a key role in the development and progression of the disease and also hypothesized that adipokines are a possible link between obesity and the other risk components of the Metabolic syndrome. Many of the adipokines exert multiple actions in a variety of cellular processes leading to a complex array of abnormal characteristic of Metabolic syndrome. Abnormal production of these adipokines by expanded visceral fat during Adiposity contributes to a pro-inflammatory state. Increasing evidence suggests that aberrant production/ release of adipokine from adipocyte i.e. adiponectin, leptin and resistin etc, may contribute to the health problems associated with Adiposity such as dyslipidemia, insulin resistance and atherosclerosis. This study conclusively have shown a significant role of adipokines secreted by adipose tissue and various metabolic risk markers play a important role in the development of Metabolic syndrome.

Keywords: Adipokine, Pathophysiology, dyslipidemia

1. Introduction

Metabolic syndrome (MetS) is a globally epidemic disorder and the term Met.S is used to describe the clustering of clinical and metabolic risk markers. It is a strong, independent contributor to the onset of coronary heart disease and type 2 diabetes (T2D) [1]. The MetS is a cluster of risk factors i.e. abdominal Adiposity, dyslipidemia, high blood pressure, insulin resistance and proinflammatory states, strongly linked to cardiovascular disease (CVD).[2,3,4,5,6,7]

2. Diagnostic criteria for metabolic syndrome

Many definitions of MetS have been laid down by the World Health Organization WHO)[9]. European Group for the Study of Insulin Resistance (EGIR) [10]. National Cholesterol Education Programme and Adult Treatment Panel III NCEP-ATP III, 2001)[11]. NCEP was slightly updated by AHA/ NHLBI in 2005 [12] and same year International Diabetes Federation (IDF) proposed a new definition [13], based on clinical criteria, which are very similar and it can be expected that they will identify many of the same individuals as having MetS. In the present study MetS was defined in accordance with the NCEP-ATP III criteria. The criteria for MetS were based on simple clinical and biochemical parameters. At least three out of five of the risk factors need to be present for clinical criteria for the MetS Whereas in case of adult Asian Indian women [14], cut-off point for waist circumference is >80 cm and body mass index (BMI) is >23 kg/m^2 (modified in various combination with components of NCEP-ATP III). Studies suggest that for people of Asian origin, as a group, the cut-off levels for the symptoms of MetS are different from non-Asians. This is one of the reasons why a single definition of the

syndrome has not yet been established and why medical organizations use “any three of the five” measures for diagnosis.

Table1:- 1. NCEP-ATP III criteria, 2001 of the metabolic syndrome

S. No.	Risk Factor	Defining Level
1.	Abdominal Adiposity, waist circumference, cm (inch)	
	Men	>102 (>40)
	Women	>88 (>35)
2.	Triglycerides, mg/dl	>150 (1.69 mmol/l)
3.	HDL cholesterol, mg/dl	
	Men	<40 (1.04 mmol/l)
	Women	<50 (1.29 mmol/l)
4.	Blood pressure, mm Hg (Systolic/Diastolic)	>130/85
5.	Fasting glucose, mg/dl	>110 (6.1 mmol/l)

3. Prevalence of Metabolic syndrome

Looking at previous studies around the world, which included general population, aged from 20-25 and upwards, the prevalence varies from 20-25% (India), 28% (USA), 30.1% (Tehran), 33.4% (Turkey) and 39.3% (Saudi Arabia). High prevalence of the Met.S has been reported from Sub-Saharan Africa and Middle East countries; South Africa, Morocco, Oman and Iran showed prevalence of 33.5%, 16.3%, 21% and 33.7%, respectively. The prevalence rates are also high in Venezuela (31.2%) and urban Brazil (25.4%) [15,16]. The situation appears to be similar in South Asian countries. MetS prevalence rates as described earlier vary among ethnic groups as defined by the NCEP-ATP III criteria among Finnish and Native American men. Both studies involved subjects

with comparable age ranges (42-60 and 44-49 years, respectively), with the Finnish study showing prevalence of only 14% compared with the prevalence in the Native American study of 43.6%. The prevalence varies from a low of 13.9% in black men to a high of 27.2% in Mexican American women [17]. These results of literature indicate that MetS is currently more prevalent and dangerous for large number of people around the world.

4. Risk markers of Metabolic syndrome

(i) Adiposity-

In fact, it is generally agreed on that increased body weight and particularly central/abdominal Adiposity is at the core of MetS [18]. It has been known for many years that body fat tends to be distributed in different patterns. That is, some people tend to store a larger amount of fat in their hips and buttocks, while others have more of their body fat in their abdomen. These patterns have in the past been termed gynoid and android Adiposity because of the general tendency of women to have more fat in peripheral sites (hips and buttocks) and men to store fat centrally (in their abdomen). However, it is clear that the distribution of body fat is much more complex than the rules of age and gender since the patterns of body fat distribution overlap considerably among men and women. It is now established that central Adiposity is associated with the various components of MetS and that Adiposity is even more predictive than total body weight of a predisposition to CVD. Central Adiposity also causes insulin resistance, a decrease of the typical responses of tissues to the hormone insulin.

(ii) Insulin resistance/Hyperinsulinemia

Insulin resistance is a central abnormality of the MetS or syndrome X or Reaven syndrome, originally hypothesized by Reaven [19], may be the link that causes the other

factors to cluster together. Insulin resistance is associated with a plethora of metabolic abnormalities (dyslipidemia, hypertension, abnormal haemostasis, low grade inflammation, altered adipokine levels) and many of these abnormalities can directly or indirectly accelerate the atherogenic process and malfunctioning of glucose metabolism. Metabolic abnormalities result from the interaction between the effects of insulin resistance located primarily in the muscle and adipose tissue. The adverse impact of the compensatory hyperinsulinaemia on tissues that remain normally insulin sensitive. [20, 21].

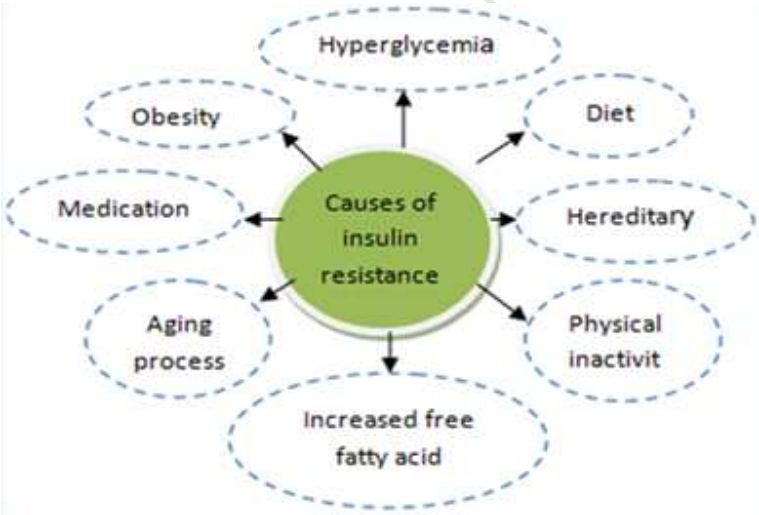


Figure:-1 Causes and consequences of insulin resistance

Fat in normal women represents between 18% and 20% of body weight, whereas in men it represents only 10% to 15%. The reason for this difference is that women at some point in their lives may nourish a fetus and then a baby from their own reserves, so women have to stock energy in the form of fat in anticipation of future pregnancies (and must stock even more energy during the last two trimesters of pregnancy). For various reasons, different fat distributions occur in women according to climate. In

hot countries, the fat is localized on the buttocks (black Africans), on the hips (Mediterraneans), and around the navel (certain Asians). This distribution avoids covering the woman with a hot coat of fat that would be difficult to bear and inefficient for thermoregulation during hot periods. In cold countries, the distribution of fat is more uniform, which provides for better protection during rigorous winters. However the fat is distributed, its main function is for the survival of the species as it provides for survival of the woman and her offspring during times of scarcity.

(iii) Diabetes

The risk for incident T2D is up to five times higher in individuals with the MetS compared with those without the syndrome [22,23], Interestingly, the presence of both the MetS and insulin resistance has an additive effect, as these patients exhibit a six to seven-fold increased risk for T2D [24,25], found that hyperinsulinemia was the strongest predictor of diabetes incidence. The presence of the MetS in women with gestational diabetes mellitus (GDM) substantially increases the risk of developing T2D. GDM alone significantly increases a woman's risk for subsequently developing T2D [26-27]. The conversion of GDM into T2D varies between 6 and 92% depending on diagnostic criteria, racial/ethnic background of the subject sample, and duration of surveillance [28]. The presence of the Met.S further increases the progression from GDM into T2D.

(iv) Dyslipidemia

Insulin resistant state-inhibited lipolysis in the adipose tissue, leading to overproduction of FFAs in the plasma and increased FFA uptake by the liver. FFA leads to increased liver concentrations of TG and cholesterol esters. High blood TG concentrations in the form of VLDL induce cholesterol ester transfer protein (CETP) activity [29], which

promotes transfer of TG from VLDL to HDL and a subsequent increase in HDL clearance and decreased HDL concentrations. It also promotes the transfer of TG into LDL in exchange for LDL cholesterol ester [30] The triglyceride-rich LDL can undergo hydrolysis by hepatic lipase or lipoprotein lipase, which leads to a small and dense, cholesterol depleted LDL particle (SD-LDL), components of atherogenic dyslipidemia are individually associated with a CVD risk [31].

(v) **Hypertension**

Hypertension is becoming a public health emergency worldwide especially in developing countries, where studies have projected up 80% increase in the numbers of hypertensive subjects by the year 2025[32] Hypertension affects approximately 1 billion individuals worldwide and is an important worldwide public-health challenge because of its high frequency and increased risks of cardiovascular and kidney disease[33,34]. It has been identified as the leading risk factor for mortality, and is ranked third as a cause of disability adjusted life-years[35]. Hypertension is a common manifestation of the metabolic disturbances associated with insulin resistance and hyperinsulinaemia, a key factor of Met.S. Some studies using the hyperinsulinaemic-euglycaemic clamp technique have demonstrated that hyperinsulinaemia occurs in hypertension as a compensatory response to reduced insulin stimulated glucose uptake by skeletal muscle [36-37]. Evidences suggest that elevations in adipocyte derived resistin and leptin may contribute to the pathogenesis of hypertension in patients with insulin resistance [38-39].

5. Pathophysiology of metabolic syndrome

Adiposity increases the risk for various co-morbidities, and recent advances in the understanding of adipose tissue biology offer an insight into the complex pathophysiologic mechanisms (Figure 2.2). Various endocrine and proinflammatory products from the visceral adipose tissue, which interact with the insulin signaling cascade, could be identified.40 Among these products, adiponectin protects from insulin resistance and cardiovascular disease[41], while free fatty acids, leptin, resistin and proinflammatory substances promote the development of insulin resistance [42]. Leptin, is primarily produced by the adipose tissue, regulates food intake and energy expenditure, fatty acid metabolism and hepatic glucose production[43]. Most common forms of Adiposity are characterized by high levels of circulating leptin and leptin resistance. Leptin reveals structural similarities with proinflammatory cytokines such as IL-6, which interferes with insulin action. Cells of the stroma vascular fraction in adipose tissue, and especially in visceral fat are an important source of IL-6 production[44]. TNF-α was extensively studied as a possible link between Adiposity, subclinical inflammation and insulin resistance. TNF-α impairs insulin action by inhibitory effects on the insulin signaling cascade and by suppressing the transcription of adiponectin.[45]. Therefore, all the adipokines are interdependent, great interest and the cross talk among them plays an important role in pathophysiology of MetS.

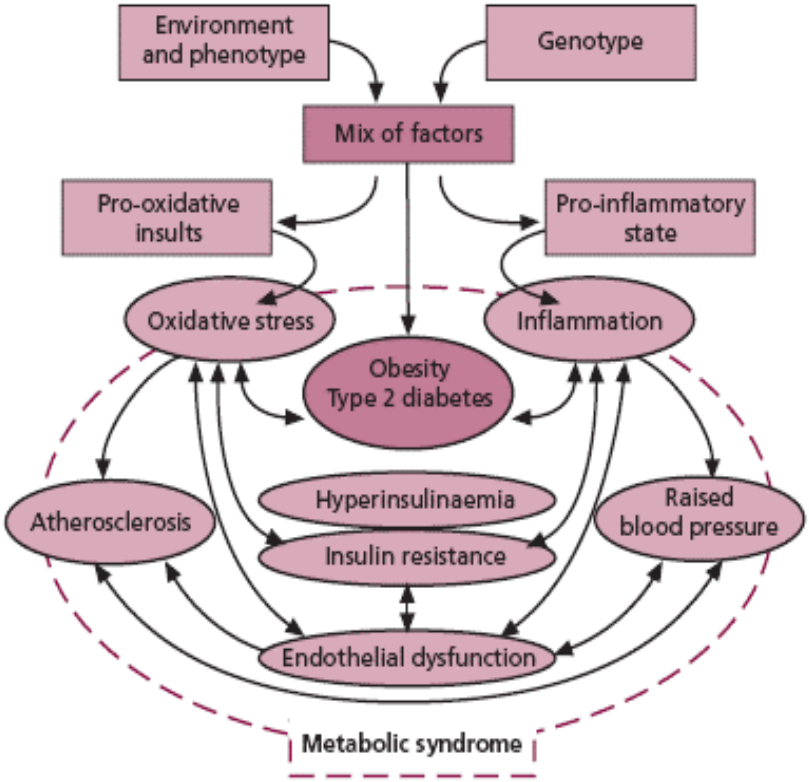


Figure 2.2: Pathophysiology of metabolic syndrome

6. Adipose tissue

Adipose tissue acts in an autocrine, paracrine or endocrine fashion [46] to control various metabolic functions and may contribute to the development of Adiposity mediated MetS. Adipose tissue functions can be classified into three aspects. First, it is related to lipid metabolism including TG storage and FA release. Second, it catabolizes TGs in order to release glycerol and FAs that participate in glucose metabolism in liver and other tissues. Finally, adipocytes secrete adipokines, which include hormones and cytokines[47] For these reasons, adipose tissue has an important influence on physiological processes such as development and growth of the adipocyte and energy

homeostasis. There are two types of adipose tissue depending on its cell structure, location, color, vascularization and function: White adipose tissue (WAT) and brown adipose tissue (BAT). WAT is the primary site of energy storage in a lipid droplet of the adipocytes in the form of TGs, whereas BAT contains multilocular adipocytes or cells with various lipid droplets. It has a large number of mitochondria and is specialized in heat production and, therefore, energy expenditure. Nevertheless, in humans, BAT is present only in newborns for regulating thermogenic process[48].

Adipose tissue contains different cell types. Only one third of the tissue is constituted by adipocytes and the rest is represented by fibroblasts, macrophages, stromal cells, monocytes and preadipocytes[49]. Hormonal activity and transcription factors are responsible for differentiation of preadipocytes to adipocytes[50]. Nevertheless, the ability of differentiation of preadipocytes is always present in all species and depends on the body energy status and the storage needs. White adipose tissue is further divided in visceral and subcutaneous adipose tissue.

(i) Subcutaneous adipose tissue (SAT)

Subcutaneous adipose tissue is located underneath the skin and is responsible for the distinct body compositions of human males and females. This type of adipose tissue contributes to temperature regulation or thermal isolation (Guyton and Hall, 2000).

(ii) Visceral adipose tissue (VAT)

Visceral adipose tissue fills in space gaps between organs and maintains them in the adequate position (Guyton and Hall, 2000). Visceral fat mass and adipocyte size is associated with peripheral and hepatic insulin resistance (Cases and Barzilai, 2000).

Although it has been demonstrated that when removing visceral fat mass, but not subcutaneous fat mass, insulin sensitivity improves [51]. it does not imply that subcutaneous adipose tissue does not contribute to several metabolic abnormalities, particularly when weight gain occurs [52] Thus, an increase in abdominal fat mass, either visceral or subcutaneous, appears to be important for the pathogenesis not only of insulin resistance but also of dyslipidemia, glucose intolerance, hypertension, and cardiovascular risk.[53,54].

7. Lipogenesis-

Lipogenesis is the synthesis of esterified FAs, which form TGs from carbohydrates or other energy sources acquired in the diet. In rats, lipogenesis occurs in liver and WAT, whereas in humans, lipogenesis contributes mildly to the fat balance. Lipid accumulation in adipose tissue depends on circulating FA uptake[55]. FAs are provided by the enzymatic hydrolysis of TG contained in the chylomicrons by the lipoprotein lipase. After FAs enter the adipocyte, reesterification is necessary for lipid storage in TG form[56]. Several enzymes involved in adipose tissue lipogenesis are induced by insulin. These are fatty acid synthase (FAS), acetyl CoA carboxylase (ACC) and malic enzyme (ME). Newly synthesized FAs are used as substrates in TG synthesis[57]. Insulin-mediated stimulation of lipogenesis in response to nutritional status is the result of an increase in the enzyme activities involved in FA biosynthesis, as well as an increase in the gene expression of these enzymes[58].

8. Lipolysis

Regional adipose differences—the degree of expansion of particular adipose tissue beds is an important factor in the development of dyslipidemia. VAT has greater

lipolytic activity than SAT, and fatty acids are directly delivered to the liver from this region via the portal vein. This leads to increased delivery of lipids to the liver and worsening IR in the liver, promoting TG synthesis and exacerbating dyslipidemia. Sam et al. found a significant link between visceral AT and dyslipidemia in patients with T2DM [59]. Expansion of VAT was associated with a greater number of VLDL and LDL particles in the circulation, even when controlling for BMI and SAT distribution. There are several differences between visceral and subcutaneous tissue, including the expression of different genes involved in insulin resistance and inflammation and, different patterns of adipokine secretion [60].

9. Adipose tissue as an endocrine organ

For many years, adipose tissue regarded merely as a heat insulator and storage of free fatty acid (FFA) that could be release when needed. However after the identification of adipokines, adipose tissue is now recognized to play a central role in the pathophysiology of insulin resistance and MetS.[61]. In recent years, it has become clear that adipose tissue is far more than a storage facility and thermoregulator and is in fact an active secretory organ of multiple mediators known as adipokines[62]. These adipokines include hormones (for example, leptin and adiponectin), inflammatory cytokines (for example, tumor necrosis factor α (TNFα), interleukin (IL)-6, omentin and visfatin) and other proteins (for example, plasminogen activator inhibitor (PAI)-1, angiotensinogen, resistin and apelin) [63].

The different fat depots (visceral vs. subcutaneous) are heterogeneous not only in terms of metabolic capacities, but also of adipokine secretion pattern. Fat depot-specific characteristics may be retained in strains derived from single human preadipocytes even

after 40 population doublings[64] .Depot-related differences in secretion may have local repercussions on the adipose tissue by autocrine or paracrine mechanisms and/or may directly impact the liver for substances produced by intraperitoneal fat. When the contribution of each depot to systemic levels of adipokines is considered, it should be kept in mind that intra-abdominal fat only represents 15% of total fat in lean and obese individuals [65].

Maintenance of a normal amount of adipose tissue is essential because imbalance can cause serious health problems and dysregulated release of adipokines may lead to metabolic disturbances and inflammation. Virtually all known adipokines are markedly dysregulated in Adiposity and Met.S. Many adipokines implicated in the pathogenesis of inflammation and insulin resistance are overproduced with increasing adiposity, thereby promoting metabolic complications.

10. Adipokines in metabolic syndrome

Many of the adipokines exert multiple actions in a variety of cellular processes leading to a complex array of abnormal characteristic of Met.S. Abnormal production of these adipokines by expanded visceral fat during Adiposity contributes to a pro-inflammatory state. Increasing evidence suggests that aberrant production/ release of adipokine from adipocyte i.e. adiponectin, leptin and resistin, may contribute to the health problems associated with Adiposity such as dyslipidemia, insulin resistance and atherosclerosis. Therefore, adipokines are potential candidates of insulin resistance, endothelial dysfunction and hypertension and reflect the importance of visceral Adiposity as a causal factor in metabolic and vascular disease.

(i) Resistin

Resistin antagonizes the action of insulin and their expression is inhibited in Adiposity and insulin resistance[66]. Elevation of the insulin and TNF-α in Adiposity, have been found to inhibit resistin expression, which may explain the low levels of resistin found in the studies on Adiposity and diabetes. A mice study suggested that resistin selectively impairs the inhibitory action of insulin on hepatic glucose production[67]. Neutralization of resistin by specific antibodies resulted in decreased blood glucose levels and improved insulin sensitivity[68], thereby providing a more direct link between fat mass and insulin resistance but the physiological role of resistin and the mechanism by which it neutralizes insulin action, are still unclear. The expression of resistin gene is induced during adipocyte differentiation of 3T3-L1 cells and the resistin polypeptide was expressed and secreted by mature adipocytes[69]. The secreted protein was found to inhibit 3T3-LI adipogenesis and it was speculated that resistin was a feedback regulator of adipogenesis. However, the role of resistin on Adiposity-associated insulin resistance has become controversial because additional evidence has suggested that Adiposity and insulin resistance are associated with decreased resistin expression.[70,71]. Resistin exists predominantly as the hexamer, although our current understanding of the metabolic effects of either complex is limited. Studies demonstrating a causal role of resistin in glucose homeostasis are based on animal models with altered serum resistin levels. Infusion or over-expression of resistin leads to hyperglycemia, a response which to a large degree can be explained by increased hepatic glucose production [72] . Despite the significant interest generated by the discovery of resistin in 2001 (Steppan et al., 2001), very little is known about the intracellular

signaling pathways by which resistin induces its metabolic effects although resistin has been shown to be important in regulating metabolic pathways in several tissues and organs including the hypothalamus, adipocytes and the liver. A consistent finding *in vivo* is that resistin suppresses liver and muscle AMPK activation [72,73,74]. however, in isolated mouse muscle this does not occur suggesting that the inhibitory effects of resistin on AMPK may require release of an unknown factor from other cell types[75].

In humans and rodents, plasma resistin concentrations are positively correlated with BMI [76]. In an animal study expression of resistin in visceral and subcutaneous adipose tissue is controversial.The present data on resistin in human samples is conflicting. Some reports have shown that resistin levels are elevated in individuals with Adiposity and diabetes[77,78,79].

(ii) Interleukin-6 (IL-6)

Interleukin-6 (IL-6) is a cytokine produced by several cells (fibroblasts, endothelial cells, monocytes, adipocytes, etc)[80].The in vivo release of IL-6 by whole-body AT could contribute to 15–35% of the systemic IL-6 in humans[81].The IL-6 receptor belongs to the class I family of cytokine receptors, which uses Janus kinases (JAKs) as intracellular signaling pathways. Circulating levels and AT production of IL-6 are increased in Adiposity [80]. IL-6 could be implicated in insulin resistance and its complications. In mice fed a high-fat diet, the increased production of IL-6 by AT induced hepatic insulin resistance.This hepatic insulin resistance could be mediated, in part, by the increased expression of suppressor of cytokine signaling (SOCS)-3, a protein that binds and inhibits the insulin receptor and also targets IRS proteins for proteosomal degradation[82],

Consistent with this report, mean plasma concentration of IL-6, but not of TNFα, leptin or MCP-1, was found to be 50% higher in the portal vein than in the radial artery of obese subjects and this portal vein IL-6 concentration correlated directly with systemic C-reactive protein concentrations[83], Also, IL-6 expression varies between adipose tissue sites: expression is higher in visceral than in peripheral adipocytes, and 90% of IL-6 expressed in adipose tissue is produced by cells other than adipocytes[84]. Overall, the association of IL-6 and insulin resistance seems complex and IL-6 alone might not be an appropriate marker of insulin resistance or Met.S.[85,86]. Fasting plasma IL-6 concentration was negatively correlated with the rate of insulin-stimulated glucose disposal in Pima Indians [87]. Circulating IL-6 level is increased in obese and insulin resistant subjects [88] and is significantly higher with greater expression in VAT[89] .

Studies in myotubes have demonstrated that these effects require activation of AMPK-activated protein kinase[90,91]. However, the exact mechanism by which IL-6 activates AMPK to promote glucose uptake and fatty acid oxidation is not yet understood. IL-6 has also helped us understand the link between Adiposity and insulin resistance. The strong correlations between adipose tissue mass and the secretion of many adipokines has led to the suggestion that reducing total adipose mass may be a strategy for the treatment of Adiposity-related diseases. However, due to the severe metabolic consequences of adipose tissue ablation as observed in lipoatrophic patients, directly modifying adipokine gene expression and release into the circulation may be more viable.

(iii)Tumor Necrosis Factor-alpha (TNF-α)

TNF-α is more expressed in visceral than in subcutaneous fat and abundantly produced by macrophages than adipocytes cells[92]. The degree of TNF-α mRNA adipose tissue expression is positively correlated with percentage body fat, body mass index, hyperinsulinemia, indicating that the amount of TNF-α present in adipose tissue may be related to insulin resistance[93], but weight loss decreased TNF-α level[94]. It significantly increases the expression of IL-6 [95], but reduces the expression of resistin in 3T3-L1 adipocyte cells[96] .It is unknown whether increased expression of TNF-α may modulate leptin expression. Neutralization of TNF-α has differential effects on critical adipokines and body composition indices; thus it improves inflammatory markers and total adiponectin in patients with the Met.S without improving insulin sensitivity. A number of studies have demonstrated that TNFα can impair insulin signaling in hepatocytes and adipose tissue [97,98]. Chronic treatment with TNFα decreases insulin-stimulated glucose uptake in rat skeletal muscle. In addition, studies using a soluble TNFα receptor-IgG chimeric protein restored insulin-induced insulin receptor and IRS-1 phosphorylation in Zucker rats in fat and skeletal muscle. Most effects of TNF α on AT are mediated by the TNFα receptor 1 subtype (TNFR1) and subsequent activation of various transduction pathways[99]. Furthermore, the targeted deletion of TNF-α or its receptors increased insulin sensitivity and glucose tolerance in obese rodents in some[100]. In obese type 2 diabetic humans, TNF-α neutralization does not appear to improve glucose tolerance or insulin sensitivity, however, in individuals without established type 2 diabetes prolonged treatment does improve insulin sensitivity[101].

Adipose tissue TNF-α concentration is correlated with Adiposity and insulin resistance in patients with and without type 2 diabetes[102]. In obese women, TNF-α messenger RNA expression in adipose tissue is correlated with fasting plasma glucose, insulin, and triacylglycerol concentration.[103]. TNF-α increases adipocyte lipolysis, and this appears to be mediated at least in part by its effects on perilipin.[104,105]. Thus, TNF-α may increase systemic insulin resistance by promoting the release of fatty acids from adipose tissue into the bloodstream to act on tissues such as muscle and liver.

(iv) **Adiponectin**

Adiponectin is a powerful inducer of proinflammatory cytokines produced by adipose tissues and macrophages[106]. The most abundant protein, adiponectin has anti-inflammatory properties affecting the nuclear factor (NF-κB) pathway and it is also a potent enhancer of insulin action on peripheral tissues[107]. The current understanding is that adiponectin acts as an endogenous insulin sensitizer by decreasing glucose levels and inducing the burning of lipid tissue in muscle and liver without increasing insulin levels. Adiponectin plasma levels are between 5–30 mg/L in lean subjects and represent 0.01% of plasma proteins. It is present in the blood stream in three main forms: trimer, hexamer and high molecular weight (HMW) 12-18-mer adiponectin.[108]. Insulin-sensitizing action of adiponectin results from AMPK-mediated reduction of hepatic gluconeogenesis and increase of muscle glucose transport. Adiponectin also enhances energy consumption and fatty acid oxidation in liver and muscle, and reduces their tissue triglyceride content; thereby further enhancing insulin sensitivity in vivo[109]. AdipoR1 and AdipoR2 serve as major receptors for adiponectin in vivo. They belong to a new family of receptors predicted to contain seven transmembrane domains but to be structurally and functionally

distinct from G-protein coupled receptors. AdipoR1 is abundantly expressed in muscle, whereas AdipoR2 is also expressed in liver [110].

Prospective studies in humans have shown that increased plasma concentrations of adiponectin are strongly and independently associated with reduced risk of incident type 2 diabetes in apparently healthy individuals and in Pima Indians[111,112]. HMW adiponectin rather than total adiponectin appears to be an important biomarker for the Met.S, being primary responsible for increased insulin sensitivity, reduced abdominal fat, high basal lipid oxidation and a more favorable lipoprotein subclass profile.Thus, lower level of HMW adiponectin could play a pathogenic role in the development of this syndrome[113].

Hypoadiponectinemia has been found in a variety of human metabolic and cardiovascular disease states including type 2 diabetes mellitus (T2DM), lipodystrophy, nonalcoholic hepatic steatosis, essential hypertension, and coronary artery disease even after body mass index (BMI) is matched. Genetic hypoadiponectinemia caused by a missense mutation has been reported. The patients carrying this mutation also exhibit a much higher propensity to develop the Met.S.[114]. As the decrease of adiponectin precedes the development of insulin resistance and myocardial infarction in humans, low levels of adiponectin are likely to be a causal component of those disorders. Adiponectin mRNA expression varies according to tissue site, being lower in visceral than in subcutaneous tissue[115]. Adiponectin also affects hepatic glucose production by decreasing the mRNA expression of two essential gluconeogenesis enzymes: phosphoenolpyruvate carboxykinase; and glucose-6-phosphatase .In addition to its effects on insulin sensitivity, adiponectin has a vascular-protective effect early in the

atherogenesis process by interfering with the regulation of adhesion molecule expression on vascular endothelial cells. the transformation of macrophages into foam cells[116], and by modulating smooth muscle cell proliferation). Also, adiponectin could reduce the inflammatory response induced by TNFα, as shown by studies in vitro where macrophage activity and TNFα production were diminished in macrophages treated with adiponectin[117].

Recently, the increase in adiponectin after weight loss was found to be correlated with serum lipid improvement, independent of insulin sensitivity changes[118]. Studies of isolated human explants or adipocytes suggest there are no differences in adiponectin secretion from visceral or subcutaneous depots[119]. Further evidence suggesting that adiponectin expression may be reduced in states of Adiposity was provided by Spiegelman and colleagues,[120] who observed decreased adiponectin mRNA in adipose tissue from obese (ob/ob) mice and humans.

This observation may represent one feature of visceral adipose tissue (VAT) biology that leads to its strong association with negative health outcomes and may also explain the finding that even after controlling for BMI and fat mass individuals with higher VAT have lower adiponectin levels than subjects with less VAT [121]. Adiponectin exhibits insulin-sensitizing, fat-burning, cardioprotective, anti-inflammatory and anti-oxidant properties, thereby thwarting several pathologies belonging to the Met.S.

(v) Leptin

Leptin is produced mainly by mature adipocytes, although production in minor extents from other tissues such as placenta, fundus of the stomach, skeletal muscle and liver has been reported[122]. Leptin concentrations have been shown to be directly

proportional to the amount of body fat. Its concentration were found to correlate with insulin, insulin resistance, glucose and measures of adiposity[123], with marked sexual dimorphism as females have higher leptin concentrations than males.

The concept of “leptin resistance” or alternate concept “hypothalamic leptin insufficiency” has been challenged and is still unclear in humans and may involve multiple mechanisms. One mechanism may be the induction of leptin on suppressor of cytokine signaling-3, which blocks the intracellular pathway of leptin[124], Another mechanism for leptin resistance may diminish the transport of leptin across the blood-brain barrier and to endoplasmic reticulum stress, which inhibit leptin signaling [125]. As body weight increased, leptin signaling protected the β-cell from adverse effects of over nutrition such as lipid accumulation, thus improving β-cell function [126], Although, leptin receptor-mediated Janus Kinase-signal transducers and activators of transcription (JAK-STAT) signaling is essential for regulation of food intake and body weight, leptin-stimulated Phosphoinositol-3-Kinase signaling appears to be important for regulation of glucose metabolism.

Circulating leptin levels parallel adipose tissue mass but also reflect immediate changes in nutritional status as they decrease soon after the beginning of fasting[127]. The secretion is also dependent on the fat depot, being higher in the subcutaneous than in the visceral depot[128]. Subcutaneous fat is responsible of 80% of total leptin production. This was shown in cultures ex vivo where the production of leptin was higher in subcutaneous adipocytes than in those of deeper origin. Leptin plasma concentration[129], and mRNA expression in adipose tissue [130] are directly related to

Adiposity severity, as an increase of fat mass is associated with an increase of leptin, which makes leptin an indicator of total fat mass.

Taken all together, these observations suggest a potentially leptin improves insulin sensitivity through activation of AMP protein kinase (AMPK), which controls cellular concentrations of malonyl-CoA, thereby inhibiting acetyl-CoA carboxylase (the enzyme involved in malonyl-CoA transformation) [131] . As a result, there is a decrease of intracellular malonyl-CoA and a decline of lipogenesis associated with increased fatty-acid beta- oxidation. Therefore, while leptin deficiency very likely contributes to insulin resistance when adipose tissue is lacking, leptin resistance is a main feature of human Adiposity. So far, the precise role of leptin in insulin resistance remains unclear. Finally we have concluded our study and found that TNF-alpha, resistin, IL-6 and Leptin levels were found higher in Metabolic syndrome populations of North India.

References:-

1. 1.Grundy SM, 2008 Metabolic syndrome pandemic. Arterioscler Thromb Vasc Biol 28: 629-636.
2. 2.Isomaa B, Almgren P, Tuomi T, Forsen B, Lahti K, Nissen M, et al, 2001Cardiovascular morbidity and mortality associated with the metabolic syndrome. Diabetes Care, 24: 683-689.
3. 3.Lakka HM, Laaksonen DE, Lakka TA, Niskanen LK, Kumpusalo E, Tuomilehto J, et al, 2002 The metabolic syndrome and total and coronary heart disease mortality in middle-aged men. The Journal of the American Medical Association. 288: 2709-2716
4. 4.Nesto RW, 2003 The relation of insulin resistance syndromes to risk of cardiovascular disease. Rev Cardiovasc Med. 4: 11-18
5. 5.Sattar N, Gaw A, Scherbakova O, Ford I, O'Reilly DS, Haffner SM ,et al, 2003 Metabolic syndrome with and without C-reactive protein as a predictor of coronary heart disease and diabetes in the West of Scotland Coronary Prevention Study. Circulation. 108: 414-419

6. 6.Katzmarzyk PT, Church TS, Blair SN, 2004 Cardiorespiratory fitness attenuates the effects of the metabolic syndrome on all-cause and coronary heart disease mortality in men. Archives of Internal Medicine, 164:1092-1097.
7. 7.Alberti KG, Zimmet PZ,2008 Should we dump the metabolic syndrome? BMJ 336: 641-643.
8. 8.DeFronzo RA, Ferrannini E,1991 Insulin Resistance. A multifaceted syndrome responsible for NIDDM, obesity, hypertension, dyslipidemia, and Atherosclerotic Cardiovascular disease. Diabetes Care. 14(3):173-194
9. 9.WHO, 1999 Definition, diagnosis and classification of diabetes mellitus and its complications. Report of a WHO consultation .
10. 10.Balkau B, Charles MA, 1999 Comment on the provisional report from the WHO consultation. European Group for the Study of Insulin Resistance (EGIR). Diabet Med. 16:442-443.
11. 11.NCEP, 2001 expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (Adult Treatment Panel III). JAMA 285: 2486–2497
12. 12.Grundy SM, Cleeman JI, Daniels SR, Donato KA, Eckel RH, Franklin BA, et al, 2005 Diagnosis and management of the metabolic syndrome: an American Heart Association/National Heart, Lung, and Blood Institute Scientific Statement. Circulation. 112: 2735-2752
13. 13.Alberti G, Zimmet P, Shaw J, 2005 The metabolic syndrome– a new worldwide definition. IDF Epidemiology Task Force Consensus Group. Lancet. 366: 1059-1062.
14. 14.Misra A, Misra R, Wijesuriya M, 2006 The metabolic syndrome in South Asians. In: Mohan V, Rao HR, Gundu HR, eds. Type 2 diabetes in South Asians. Epidemiology, risk factors and prevention. New Delhi: Jaypee Brothers. 76–96
15. 15.Florez H, Silva E, Fernandez V, Ryder E, Sulbaran T, Campos G, et al, 2005 Prevalence and risk factors associated with the metabolic syndrome and dyslipidaemia in White, Black, Amerindian and Mixed Hispanics in Zulia State, Venezuela. Diabetes Res Clin Pract. 69:63–77
16. 16.Marquezine GF, Oliveira CM, Pereira AC, Krieger JE, Mill JG, 2008 Metabolic syndrome determinants in an urban population from Brazil: social class and gender-specific interaction. Int J Cardiol. 129: 259–265.
17. 17.Okosun IS, Liao Y, Rotimi CN, Prewitt TE, Cooper RS, 2000 Abdominal Obesity and clustering of multiple factors in metabolic syndrome in White, Black and Hispanic Americans. Ann Epidemiol. 10: 263-270.
18. 18.Phillips SA, Ciaraldi,TP, Oh DK, Savu MK, and Henry RR, 2008 Adiponectin secretion and response to pioglitazone is depot dependent in cultured human

adipose tissue. Am J Physiol Endocrinol Metab 295: 842-850.

19. 19.Reaven GM, 1988 Role of insulin resistance in human disease. Diabetes 37: 1595–1607.

20. 20.Reaven GM, 2001 Insulin resistance, compensatory hyperinsulinemia, and coronary heart disease: syndrome X revisited. In Handbook of Physiology, Section 7 The Endocrine System, Vol. II The Endocrine Pancreas and Regulation of Metabolism. LS Jefferson, AD Cherrington, (eds). New York, Oxford University Press, 1169-1197.

21. 21.Reaven GM, 2004 Insulin resistance and its consequences. In Diabetes Mellitus: A Fundamental and Clinical Text. D LeRoith, SI Taylor, JM Olefasky (eds). Philadelphia, Lippincott, Williams and Wilkins, 899-915

22. 22.Grundy SM, 2006 Metabolic syndrome: connecting and reconciling cardiovascular and diabetes worlds. J Am Coll Cardiol 47: 1093-1100.

23. 23.Ford ES, Li C, Sattar N, 2008 Metabolic Syndrome and Incident Diabetes: Current state of the evidence. Diabetes Care 31: 1898-1904.

24. 24.Meigs JB, Rutter MK, Sullivan LM, Fox CS, D'Agostino RB, Sr Wilson PW, 2007 Impact of insulin resistance on risk of type 2 diabetes and cardiovascular disease in people with metabolic syndrome. Diabetes Care 30: 1219-1225.

25. 25.Hanson RL, Imperatore G, Bennett PH, Knowler WC, 2002 Components of the "metabolic syndrome" and incidence of type 2 diabetes. Diabetes 51: 3120-3127.

26. 26.Pyorala M, Miettinen H, Halonen P, Laakso M, Pyorala K, 2000 Insulin resistance syndrome predicts the risk of coronary heart disease and stroke in healthy middle-aged men: the 22-year follow-up results of the Helsinki Policemen Study. Arterioscler Thromb Vasc Biol 20: 538-544.
27. 27.Jovanovic L, Pettitt DJ, 2001 Gestational diabetes mellitus. JAMA 286: 2516-2518.

28. 28.Kim C, Newton KM, Knopp RH, 2002 Gestational diabetes and the incidence of type 2 diabetes: a systematic review. Diabetes Care 25: 1862-1868.

29. 29.Murakami T, Michelagnoli S, Longhi R, Gianfranceschi G, Pazzucconi F, Calabresi L, et al, 1995 Triglycerides are major determinants of cholesterol esterification/transfer and HDL remodeling in human plasma. Arterioscler Thromb Vasc Biol 15: 1819-1828.
30. 30.Austin MA, King MC, Vranizan KM, Krauss RM, 1990 Atherogenic lipoprotein phenotype. A proposed genetic marker for coronary heart disease risk. Circulation 82: 495-506.
31. 31.Caballero AE, 2004 Endothelial dysfunction, inflammation and insulin resistance: a focus on subjects at risk for type 2 diabetes. Curr Diab Rep. 4: 237-246

32. 32.Kearney PM, Whelton M, Reynolds K, Whelton PK, He J, 2004 Worldwide prevalence of hypertension: a systematic review. J Hypertens 22: 11–19.
33. 33.Whelton PK,1994 Epidemiology of hypertension. Lancet 344: 101–106.
34. 34He J, Whelton PK, 1997 Epidemiology and prevention of hypertension. Med Clin North Am 81: 1077–1097
35. 35.Ezzati M, Lopez AD, Rodgers A, Vander Hoorn S, Murray CJ, 2002 Selected major risk factors and global and regional burden of disease. Lancet. 360: 1347–1360
36. 36.Katsilambros N, Aliferis K, Darviri C, Tsapogas P, Alexiou Z, Tritos N, et al,1993 Evidence for an increase in the prevalence of known diabetes in a sample of an urban population in Greece. Diabet Med. 10: 87–90.
37. 37.Ramachandran A, Snehalatha C, Latha E, Vijay V, Viswanathan M,1997 Rising prevalence of NIDDM in an urban population in India. Diabetologia 40: 232–237
38. 38.Bernal-Mizrachi C, Weng S, Li B, Nolte LA, Feng C, Coleman T, 2002 Respiratory uncoupling lowers blood pressure through a leptin dependent mechanism in genetically obese mice. Arterioscler Thromb Vasc Biol 22: 961-968
39. 39.Takata Y, Osawa H, Kurata M, Kurokawa M, Yamauchi J, Ochi M, 2008 Hyperresistinemia is associated with coexistence of hypertension and type 2 diabetes. Hypertension 51: 534-539
40. 40.Havel PJ, 2004 Update on adipocyte hormones; regulation of energy balance and carbohydrate/ lipid metabolism. Diabetes 53: 143–151
41. 41.Matsuzawa Y, Funahashi T, Kimara S, Shimomura I,2004 Adiponectin and metabolic syndrome. Arterioscler Thromb Vasc Biol 24: 29–33
42. 42.Ridker PM, 2007 Inflammatory biomarkers and risk for myocardial infarction, stroke, diabetes and total mortality: implications for longevity. Nutr Rev. 65: S253–S259
43. 43.Dyck DJ, Heigenhauser GJ, Bruce CR, 2006 The role of adipokines as regulators of skeletal muscle fatty acid metabolism and insulin sensitivity. Acta Physiol. 186: 5–16

44. 44.Grimble RF, 2002 Inflammatory status and insulin resistance. Curr Opin Clin Nutr Metab Care 5: 551.
45. 45.Hotamisligil GS, Arner P, Caro JF, Atkinson RL, Spiegelman BM 1995 Increased adipose tissue expression of tumor necrosis factor alpha in human obesity and insulin resistance. J Clin Invest. 95: 2409–2415.
46. 46.Ahima RS, Flier JS, 2000. Adipose tissue as an endocrine organ. Trends Endocrinol Metab 11:327-332.

47. 47.Morrison JA, Friedman LA, Wang P, Glueck CJ, 2008 Metabolic syndrome in childhood predicts adult metabolic syndrome and type 2 diabetes mellitus 25 to 30 years later. J Pediatr 152: 201-206.

48. 48.Gesta S Tseng, YH Kahn, CR.,2007 Developmental origin of fat: Tracking obesity to its source. Cell 131:242–256.

49. 49Geloen A, Collet A J, Guay G, Bukowiecki L J, 1989 Insulin stimulates *in vivo* cell proliferation in white adipose tissue. Am. J. Physiol. 256, C190–C196.GEMS Study. Obesity 17: 737–744.

50. 50.Farmer SR, 2006 Transcriptional control of adipocyte formation. Cell Metab 4: 263-273.

51. 51 FH, Atzmon G, Yang XM, Ma XH, Rincon M, Rudin E, 2005 Differential responses of visceral and subcutaneous fat depots to nutrients. Diabetes 54: 672-678.

52. 52.Smith SR, Lovejoy JC, Greenway F, Ryan D, deJonge L, de la Bretonne J, 2001). Contributions of total body fat, abdominal subcutaneous adipose tissue compartments, and visceral adipose tissue to the metabolic complications of obesity. Metabolism 50: 425-435

53. 53Jensen M D, 2006 Is visceral fat involved in the pathogenesis of the metabolic syndrome? Human model. Obesity (Silver Spring) 1: 20S-24S.

54. 54Stenvinkel P, 2000 Leptin and blood pressure--is there a link? Nephrol Dial Transplant 15:1115-1117.

55. 55R Zechner, J Strauss , S Frank , E Wagner , W Hofmann , D Kratky , 2000 The role of lipoprotein lipase in adipose tissue development and metabolism International Journal of Obesity .24:S53±S56.

56. 56Hirata K, Dichek H L, Cioffi J A, Choi S Y, Leeper N J, Quintana L, 1999 Cloning of a unique lipase from endothelial cells extends the lipase gene family. J Biol Chem 274: 14170-14175.

57. 57Sul H S, Wang D, 1998 Nutritional and hormonal regulation of enzymes in fat synthesis: studies of fatty acid synthase and mitochondrial glycerol-3- phosphate acyltransferase gene transcription. Annu. Rev. Nutr. 18: 331–351.

58. 58Va´zquez-Vela ME, Torres N, Tovar A R 2008 White adipose tissue as endocrine organ and its role in obesity. Arch. Med. Res. 39: 715–728.

59. 59Sam S, Haffner S, Davidson MH, D'Agostino RB, Feinstein S, Kondos G, 2008 Relationship of abdominal visceral and subcutaneous adipose tissue with lipoprotein particle number and size in type 2 diabetes. Diabetes 57: 2022-2027.

60. 60Lafontan M, Berlan M, 2003 Do regional differences in adipocyte biology provide new pathophysiological insights? Trends Pharmacol Sci 24 :276-283.

61. 61 C, Foley J E, Bogardus C, Tataranni PA, Pratley RE 2000 Enlarged subcutaneous abdominal adipocyte size, but not obesity itself, predicts type II diabetes independent of insulin resistance. Diabetologia 43: 1498-1506.

62. 62Mohamed-Ali V, Pinkney JH, Coppack SW, 1998 Adipose tissue as an endocrine and paracrine organ. Int J Obes Relat Metab Disord 22: 1145-1158.

63. 63Trayhurn P, Wood I S, 2004 Adipokines: inflammation and the pleiotropic role of white adipose tissue. Br J Nutr 92: 347-355.

64. 64Tchkonia T, Giorgadze N, Pirtskhalava T, Thomou T, DePonte M, Koo A., et al,2006 Fat depot-specific characteristics are retained in strains derived from single human preadipocytes. Diabetes 55: 2571-2578.

65. 65 Klein S, Allison D B, Heymsfield S B, Kelley D E, Leibel R L, Nonas C, 2007 Waist circumference and cardiometabolic risk: a consensus statement from shaping America's health: Association for Weight Management and Obesity Prevention; NAASO, the Obesity Society; the American Society for Nutrition;

and the American Diabetes Association. Diabetes Care 30, 1647-1652.

66. Way JM, Görgün CZ, Tong Q, Uysal KT, Brown K K, Harrington W W, Oliver, 2001 Adipose tissue resistin expression is severely suppressed in obesity and stimulated by peroxisome proliferator-activated receptor gamma agonists. J Biol Chem 276: 25651-25653.
67. Rajala MW, Qi Y, Patel HR, Takahashi N, Banerjee R, Pajvani UB, 2004 Regulation of resistin expression and circulating levels in obesity, diabetes, and fasting. Diabetes 53: 1671-1679.
68. Steppan CM, Bailey ST, Bhat S, Brown EJ, Banerjee RR, Wright CM, et al, 2001 The hormone resistin links obesity to diabetes. Nature 409:307–312.
69. Steppan CM, Lazar MA, 2002 Resistin and obesity-associated insulin resistance. Trends Endocrinol Metab 13: 18-23.
70. Juan CC, Au LC, Fang VS, Kang SF, Ko YH, Kuo SF, 2001 Suppressed gene expression of adipocyte resistin in an insulin-resistant rat model probably by elevated free fatty acids. Biochem Biophys Res Commun. 289: 1328–1333.
71. Milan G, Granzotto M, Scarda A, Calcagno A, Pagano C, Federspil G,et al, 2002 Resistin and adiponectin expression in visceral fat of obese rats: effect of weight loss. Obes Res 10: 1095–1103
72. Banerjee R R, Rangwala SM, Shapiro J S, Rich A S, Rhoades B, Qi Y, et al, 2004 Regulation of fasted blood glucose by resistin. Science 303:1195-1198.
73. Qi Y, Nie Z, Lee YS, Singhal NS, Scherer PE., Lazar MA, and Ahima RS, 2006 Loss of resistin improves glucose homeostasis in leptin deficiency. Diabetes 55: 3083-3090.
74. Satoh H, Nguyen MT, Miles PD, Imamura T, Usui I, Olefsky JM, 2004 Adenovirus-mediated chronic "hyper-resistinemia" leads to in vivo insulin resistance in normal rats. J Clin Invest 114: 224-231
75. Jorgensen SB, Honeyman J, Oakhill JS, Fazakerley D, Stockli J, Kemp BE, et al, 2009 Oligomeric resistin impairs insulin and AICAR-stimulated glucose uptake in mouse skeletal muscle by inhibiting GLUT4 translocation. American journal of physiology Endocrinology and metabolism. 297:E57–66.
76. Hui-Ling Lu, Hong-Wei Wang, Yu Wen, Mu-Xun Zhang, Han-Hua Lin, 2006 Roles of adipocyte derived hormone adiponectin and resistin in insulin resistance of type 2 diabetes World J Gastroenterol. 12(11): 1747-1751.
77. Fujinami A, Obayashi H, Ohta K, Ichimura T, Nishimura M, Matsui H, 2004 Enzyme-linked immunosorbent assay for circulating human resistin: resistin concentrations in normal subjects and patients with type 2 diabetes. Clin Chim Acta 339: 57-63
78. Youn BS, Yu KY, Park HJ, Lee NS, Min SS, Youn MY, et al, 2004 Plasma resistin concentrations measured by enzyme-linked immunosorbent assay using a newly developed monoclonal antibody are elevated in individuals with type 2 diabetes mellitus. J Clin Endocrinol Metab 89:150-156.

79. Degawa-Yamauchi M, Bovenkerk JE, Juliar BE, Watson W, Kerr K, Jones R, 2003 Serum resistin (FIZZ3) protein is increased in obese humans. J Clin Endocrinol Metab 88: 5452-5455.
80. Bastard JP, Lagathu C, Caron M, Capeau J, 2007 Point-counterpoint: interleukin-6 does/does not have a beneficial role in insulin sensitivity and glucose homeostasis. J. Appl. Physiol. 102: 821–822.
81. Mohamed-Ali V, Goodrick S, Rawesh A, Katz DR, Miles JM, Yudkin JS, 1997 Subcutaneous adipose tissue releases interleukin-6, but not tumor necrosis factor-α, in vivo. J Clin Endocrinol Metab. 82: 4196–4200.
82. Sabio G, Das M, Mora A, Zhang Z, Jun JY, Ko HJ, 2008 A stress signaling pathway in adipose tissue regulates hepatic insulin resistance. Science 322: 1539-1543
83. Fontana L, Eagon JC, Trujillo ME, Scherer PE, Klein S, 2007 Visceral fat adipokine secretion is associated with systemic inflammation in obese humans. Diabetes 56: 1010-1013
84. Fried SK, Bunkin DA, Greenberg AS ,1998 Omental and subcutaneous adipose tissues of obese subjects release interleukin-6: depot difference and regulation by glucocorticoid. J Clin Endocrinol Metab. 83: 847–850.
85. Salmenniemi U, Ruotsalainen E, Pihlajamäki J, Vauhkonen I, Kainulainen , Punnonen K, 2004 Multiple abnormalities in glucose and energy metabolism and coordinated changes in levels of adiponectin, cytokines, and adhesion molecules in subjects with metabolic syndrome. Circulation 110: 3842-3848.
86. Matsushita K, Yatsuya H, Tamakoshi K, Wada K, Otsuka R, Takefuji S, 2006 Comparison of circulating adiponectin and proinflammatory markers regarding their association with metabolic syndrome in Japanese men. Arterioscler Thromb Vasc Biol 26: 871-876.
87. Vozarova B, Weyer C, Hanson K, Tataranni PA, Bogardus C, Pratley RE, 2001 Circulating interleukin-6 in relation to adiposity, insulin action, and insulin secretion. Obes Res. 9: 414–417
88. Kern PA, Ranganathan S, Li C, Wood L, Ranganathan G, 2001 Adipose tissue tumor necrosis factor and interleukin-6 expression in human obesity and insulin resistance. Am J Physiol Endocrinol Metab 280: E745-751
89. Okuno A, Tamemoto H, Tobe K, Ueki K, Mori Y, Iwamoto K, 1998 Troglitazone increases the number of small adipocytes without the change of white adipose tissue mass in obese Zucker rats. J Clin Invest 101: 1354-1361.
90. Carey AL, Steinberg GR, Macaulay SL, Thomas WG, Holmes AG, Ramm G, 2006 Interleukin-6 increases insulin-stimulated glucose disposal in humans and glucose uptake and fatty acid oxidation in vitro via AMP-activated protein kinase. Diabetes 55: 2688-2697.
91. Al-Khalili L, Bouzakri K, Glund S, Lönnqvist F, Koistinen HA, Krook A, 2006

Signaling specificity of interleukin-6 action on glucose and lipid metabolism in skeletal muscle. Mol Endocrinol. (12):33 64-75

92. Maury E, Noël L, Detry, Brichard SM, 2009 In vitro hyperresponsiveness to tumor necrosis factor-alpha contributes to adipokine dysregulation in omental adipocytes of obese subjects. J Clin Endocrinol Metab 94: 1393-1400.
93. Hotamisligil GS, Arner P, Caro JF, Atkinson RL, Spiegelman BM (1995). Increased adipose tissue expression of tumor necrosis factor alpha in human obesity and insulin resistance. J Clin Invest. 95: 2409–2415.
94. Jellema A, Plat J, Mensink RP, 2004 Weight reduction, but not a moderate intake of fish oil, lowers concentrations of inflammatory markers and PAI-1 antigen in obese men during the fasting and postprandial state. Eur. J. Clin. Invest. 34: 766–773.
95. Fasshauer M, Klein J, Lossner U, Paschke R, 2003 Interleukin-6 mRNA expression is stimulated by insulin, isoproterenol, tumour necrosis factor alpha, growth hormone, and IL-6 in 3T3-L1 adipocytes. Horm Metab Res. Mar; 35 (3): 147-152.
96. Fasshauer M, Klein J, Neumann S, Eszlinger M, Paschke R, 2001 Tumor necrosis factor alpha is a negative regulator of resistin gene expression and secretion in 3T3-L1 adipocytes. Biochem Biophys Res Commun. Nov 9; 288 (4): 1027-1031.
97. Ruan H, Miles PD, Ladd CM, Ross K, Golub TR, Olefsky JM, 2002 Profiling gene transcription in vivo reveals adipose tissue as an immediate target of tumor necrosis factor-alpha: implications for insulin resistance. Diabetes 51: 3176-3188
98. Cai D, Yuan M, Frantz DF, Melendez PA, Hansen L, Lee J, 2005 Local and systemic insulin resistance resulting from hepatic activation of IKK-beta and NF-kappaB. Nat Med 11: 183-190
99. Cawthorn, W.P., and Sethi, J.K. 2008 TNF-alpha and adipocyte biology. FEBS Lett 582: 117-131.
100. Uysal, K.T., Wiesbrock, S.M., Marino, M.W., and Hotamisligil, G.S. (1997). Protection from obesity-induced insulin resistance in mice lacking TNF-alpha function. Nature 389, 610-614.

101. Tam LS, Tomlinson B, Chu TT, Li T, Li EK, 2007 Impact of TNF inhibition on insulin resistance and lipids levels in patients with rheumatoid arthritis. Clin Rheumatol 26:1495-1498.
102. Hotamisligil GS, Spiegelman BM, 1994 Tumor necrosis factor alpha: a key component of the obesity-diabetes link. Diabetes 43: 1271-1278.
103. Hotamisligil GS, Shargill NS, Spiegelman B M, 1993 Adipose expression of tumor necrosis factor-alpha: direct role in obesity-linked insulin resistance. Science 259: 87-91
104. Zhang H H, Halbleib M, Ahmad F, Manganiello VC, Greenberg AS,

2002 Tumor necrosis factor-alpha stimulates lipolysis in differentiated human adipocytes through activation of extracellular signal-related kinase and elevation of intracellular cAMP. Diabetes 51: 2929-2935.

105. Souza SC, Palmer HJ, Kang YH, Yamamoto MT, Muliro KV, Paulson KE, 2003 TNF-alpha induction of lipolysis is mediated through activation of the extracellular signal related kinase pathway in 3T3-L1 adipocytes. J Cell Biochem 89: 1077-1086.

106. Tsatsanis C, Zacharioudaki V, Androulidaki A, Dermitzaki E, Charalampopoulos I, Minas V, 2006 Peripheral factors in the metabolic syndrome: the pivotal role of adiponectin. Ann N Y Acad Sci 1083: 185-195

107. Stefan N, Bunt JC, Salbe AD, Funahashi T, Matsuzawa Y, Tataranni PA, 2002 Plasma adiponectin concentrations in children: relationships with obesity and insulinemia. J Clin Endocrinol Metab 87: 4652-4656

108. Kadowaki H, Hagura R, Akanuma Y, Yazaki Y, Nagai R, Taniyama M, Matsubara K, 2002 Genetic variation in the gene encoding adiponectin is associated with an increased risk of type 2 diabetes in the Japanese Population. Diabetes. 51: 536-540.

109. Kadowaki T, 2001 The fat-derived hormone adiponectin reverses insulin resistance associated with both lipoatrophy and obesity. Nat Med 7: 941–946.

110. Yamauchi T, Kamon J, Minokoshi Y, Ito Y, Waki H, Uchida S, Yamashita, S, 2002 Adiponectin stimulates glucose utilization and fatty-acid oxidation by activating AMP-activated protein kinase. Nat Med 8: 1288-1295

111. Lindsay RS, Funahashi T, Hanson RL, Matsuzawa Y, Tanaka S, Tataranni PA, 2002 Adiponectin and development of type 2 diabetes in the Pima Indian population. Lancet 360:57-58.

112. Spranger J, Kroke A, Möhlig M, Bergmann MM, Ristow M, Boeing H, 2003 Adiponectin and protection against type 2 diabetes mellitus. Lancet 361: 226-228.

113. Heidemann C, Sun Q, van Dam RM, Meigs JB, Zhang C, Tworoger SS, et al, 2008 Total and high-molecular-weight adiponectin and resistin in relation to the risk for type 2 diabetes in women. Ann Intern Med 149: 307-316

114. Kondo H, Shimomura I, Matsukawa Y, Kumada M, Takahashi M, Matsuda M, Ouchi N, 2002 Association of adiponectin mutation with type 2 diabetes: a candidate gene for the insulin resistance syndrome. Diabetes 51: 2325-2328.

115. Lihn AS, Bruun JM, He G, Pedersen SB, Jensen PF, Richelsen B, 2004 Lower expression of adiponectin mRNA in visceral adipose tissue in lean and obese subjects. Mol Cell Endocrinol 219: 9-15.

116. Ouchi N, Kihara S, Arita Y, Maeda K, Kuriyama H, Okamoto Y, Hotta K, Nishida M,et al, 2001 Adipocyte-derived plasma protein, adiponectin, suppresses

lipid accumulation and class A scavenger receptor expression in human monocyte-derived macrophages. Circulation 103: 1057-1063.

117. Ouchi N, Kihara S, Arita Y, Okamoto Y, Maeda K, Kuriyama H, Hotta Z, et al, 2000 Adiponectin, an adipocyte-derived plasma protein, inhibits endothelial NF-kappaB signaling through a cAMP-dependent pathway. Circulation 102: 1296-1301.

118. Baratta R, Amato S, Degano C, Farina MG, Patanè G, Vigneri R, 2004 Adiponectin relationship with lipid metabolism is independent of body fat mass: evidence from both cross-sectional and intervention studies. J Clin Endocrinol Metab. 89: 2665-2671

119. Fain JN, Bahouth SW, Madan AK, 2004 TNFalpha release by the nonfat cells of human adipose tissue. Int J Obes Relat Metab Disord 28: 616-622.

120. Hu E, Liang P, Spiegelman BM, 1996 AdipoQ is a novel adipose-specific gene dysregulated in obesity. J Biol Chem 271: 10697-10703.

121. Bacha F, Saad R, Gungor N, Arslanian SA, 2004 Adiponectin in youth: relationship to visceral adiposity, insulin sensitivity, and beta-cell function. Diabetes Care. 27: 547–552.

122. Zhang Y, Proenca R, Maffei M, Barone M, Leopold L, Friedman JM, 1994 Positional cloning of the mouse obese gene and its human homologue. Nature 372: 425-432.

123. Jazet IM, Pijl H, Meinders AE, 2003 Adipose tissue as an endocrine organ: impact on insulin resistance. Neth J Med. 61(6): 194-212.

124. Munzberg H, Myers MG Jr, 2005 Molecular and anatomical determinants of central leptin resistance. Nat. Neurosci. 8: 566–570.

125. Ozcan L, Ergin AS, Lu A, Chung J, Sarkar S, Nie D, et al, 2009 Endoplasmic reticulum stress plays a central role in development of leptin resistance. Cell Metab 9:35-51.

126. Morioka T, Asilmaz E, Hu J, Dishinger JF, Kurpad AJ, Elias CF, 2007 Disruption of leptin receptor expression in the pancreas directly affects beta cell growth and function in mice. J. Clin. Invest. 117: 2860–2868.

127. Becker DJ, Ongemba LN, Brichard V, Henquin JC, Brichard SM, 1995 Diet- and diabetes-induced changes of ob gene expression in rat adipose tissue. FEBS Lett 371: 324-328

128. Montague CT, Prins JB, Sanders L, Zhang J, Sewter CP, Digby J, 1998 Depot-related gene expression in human subcutaneous and omental adipocytes. Diabetes 47:1384-1391.

129. Considine RV, Sinha MK, Heiman ML, Kriauciunas A, Stephens TW, Nyce MR,1996 Serum immunoreactive-leptin concentrations in normal-weight and obese humans. N Engl J Med 334: 292-295.

130. Vidal H, Auboeuf D, De Vos P, Staels B, Riou JP, Auwerx J, 1996 The

expression of ob gene is not acutely regulated by insulin and fasting in human abdominal subcutaneous adipose tissue. J Clin Invest 98: 251-255.

131. Minokoshi Y, Kim YB, Peroni OD, Fryer LG, Muller C, Carling D, 2002 Leptin stimulates fatty-acid oxidation by activating AMP-activated protein kinase. Nature. 415: 339–343.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Conflict of Interest:- None

ACCEPTED MANUSCRIPT